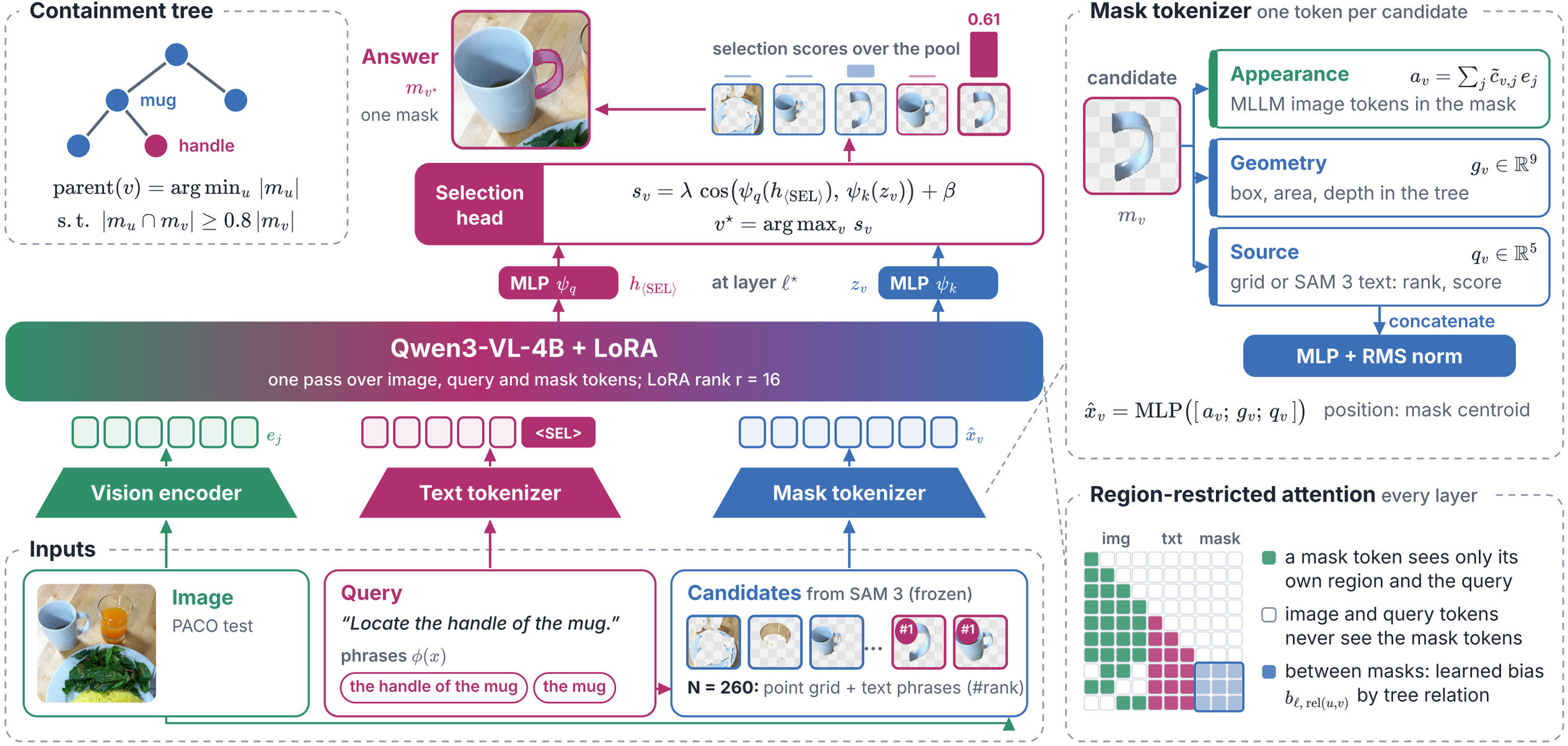

Containment tree
mug
handle
$\text{parent}(v) = \arg\min_u |m_u|$
s. t. $|m_u \cap m_v| \geq 0.8\,|m_v|$
Answer
$m_{v^\star}$
one mask
selection scores over the pool
0.61
Selection head
$s_v = \lambda \cos\big(\psi_q(h_{\langle \text{SEL} \rangle}),\, \psi_k(z_v)\big) + \beta$
$v^\star = \arg\max_v s_v$
MLP ψ_q
$h_{\langle \text{SEL} \rangle}$
at layer $\ell^\star$
z_v
MLP ψ_k
Qwen3-VL-4B + LoRA
one pass over image, query and mask tokens; LoRA rank r = 16
e_j
<SEL>
$\hat{x}_v$
Vision encoder
Text tokenizer
Mask tokenizer
Inputs
Image
PACO test
Query
"Locate the handle of the mug."
phrases $\phi(x)$
the handle of the mug
the mug
Candidates from SAM 3 (frozen)
#1
#1
N = 260: point grid + text phrases (#rank)
Mask tokenizer one token per candidate
candidate
m_v
Appearance
$a_v = \sum_j \tilde{c}_{v,j}\, e_j$
MLLM image tokens in the mask
Geometry
$g_v \in \mathbb{R}^9$
box, area, depth in the tree
Source
$q_v \in \mathbb{R}^5$
grid or SAM 3 text: rank, score
concatenate
MLP + RMS norm
$\hat{x}_v = \text{MLP}\big([\,a_v;\, g_v;\, q_v\,]\big)$
position: mask centroid
Region-restricted attention every layer
img
txt
mask
a mask token sees only its own region and the query
image and query tokens never see the mask tokens
between masks: learned bias $b_{\ell,\,\text{rel}(u,v)}$ by tree relation